Alkenes

Exam code: 0620 & 0971

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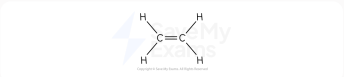
Catalytic cracking

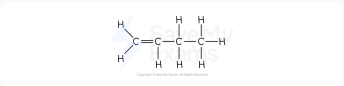
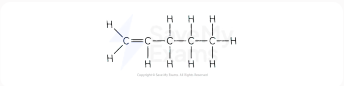
- Alkenes are **unsaturated** hydrocarbons
  - This means that they contain at least one carbon-carbon double bond (C=C)
- They have **covalent** bonds
- Their general formula is $\mathbf{C_nH_{2n}}$
- The presence of the double bond, C=C, means they can **make more bonds** with other atoms by opening up the C=C bond and allowing incoming atoms to form another single bond with each carbon atom of the functional group
- Each of these carbon atoms now forms 4 single bonds instead of 1 double and 2 single bonds
- This makes them much more reactive than alkanes

Table of alkenes

Displayed formula	Name	Molecular formula
	ethene	C_2H_4

Help me understand

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

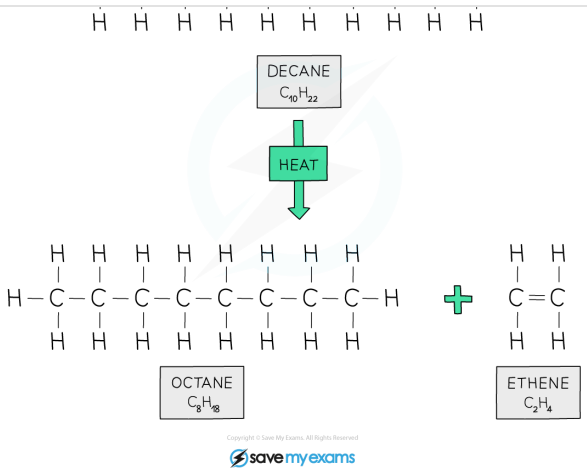
	but-1-ene	C_4H_8
	pent-1-ene	C_5H_{10}

The first four members of the alkene homologous series

Manufacture of alkenes

- Although there is use for each fraction obtained from the fractional distillation of crude oil, the amount of longer chain hydrocarbons produced is far greater than needed
- These **long chain hydrocarbon** molecules are further processed to produce other products
- A process called **catalytic cracking** is used to convert longer-chain molecules into **short-chain** and more useful hydrocarbons
- Shorter chain alkanes, alkenes and hydrogen are produced from the cracking of longer chain alkanes
- Alkenes can be used to make polymers and the hydrogen used to make ammonia
- Kerosene and diesel oil are often cracked to produce petrol, other alkenes and hydrogen
- **Cracking** involves heating the hydrocarbon molecules to around 600 – 700°C to **vaporise** them
- The vapours then pass over a hot powdered **catalyst** of alumina or silica
- This process breaks covalent bonds in the molecules as they come into contact with the surface of the catalyst, causing **thermal decomposition** reactions
- The molecules are broken up in a random way which produces a mixture of smaller alkanes and alkenes
- Hydrogen and a higher proportion of alkenes are formed at higher temperatures and higher pressure
-

The cracking of decane

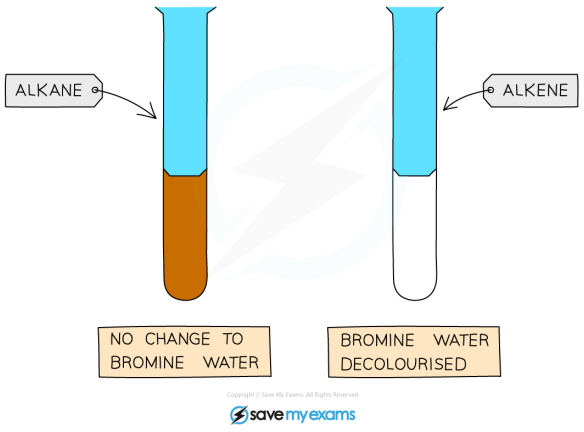


Decane is catalytically cracked to produce octane for petrol and ethene for ethanol

Distinguishing from alkanes

- The presence of the C=C double bond in an alkene but not an alkane allows us to distinguish between them
- To do this, **bromine water** is used
- Bromine water is an orange coloured solution
- When bromine water is shaken with an alkane the solution **remains orange**
- When bromine water is shaken with an alkene, the solution will go **colourless,** as the bromine can add across the double bond meaning it is no longer in solution

How to distinguish between alkanes and alkenes



Alkenes will decolourise bromine water

ensure you say colourless, and not clear.

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